STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. What measures translational inertia? [1]
2. Second law at constant mass? [1]
3. Do action-reaction forces act on one body? [1]
4. Momentum formula? [1]
5. Net force at equilibrium? [1]
6. Kinetic-friction direction? [1]
7. What provides centripetal force? [1]
8. Maximum static friction? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. 5 kg under net force 20 N: a? [2]
10. 0.2 kg changes velocity +10 to -5 m/s: Deltap? [2]
11. 10 kg resting on level floor, g=9.8: N? [2]
12. 2 kg at 6 m/s in radius 3 m: force? [2]
13. 50 N for .2 s: impulse? [2]
14. Why does swimmer push water backwards? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. 6 kg trolley pulled 30 N, friction 6 N: a? [3]
16. 0.15 kg ball at 20 m/s stops in .05 s: average force? [3]
17. 10 kg block on frictionless 30 deg incline, g=10: a,N? [3]
18. 2 kg at 5 m/s in radius 2.5 m: radial force? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. 4 kg block released on frictionless 30 deg incline: speed after 3 s, g=10? [4]
20. 0.5 kg ball rebounds +12 to -8 m/s in .1 s: average force? [4]

End of paper - check working, units, signs and significant figures.